

# AudioPlatform

*for Teensy 4.1*

## Getting Started Guide



**Revision 1.0.1, August 8, 2025**

***Donald Swearingen and ProtoSupplies***

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# Welcome

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First and foremost, AudioPlatform for Teensy 4.1 is a hardware platform for experimentation and development of simple-to-sophisticated, stand-alone, audio systems and applications, built within and upon the Teensy4.1/TeensyDuino/Arduino environment.

AudioPlatform is a response to the difficulty that many potential creators of Teensy-based audio systems face: the challenge of acquiring or themselves building some sort of hardware, often a “DIY” PCB board and list of components, to accommodate their projects and explorations. Many of those with such interests simply lack the skills or resources to complete this process; and the general lack of availability of off-the-shelf hardware suitable to their projects restricts their options to the use of breadboards and/or simply-wired “perf” boards to do so, not to mention creating modern PCB-based devices with surface mounted components.

This was pretty much the case for me, as well. Although I’m a life-long professional musician and composer, I’ve also over the years acquired considerable skills in software and electronic systems development, creating software and hardware components and systems in support of numerous live musical and artistic projects. Most of that work had also involved the use of powerful computers.

AudioPlatform aims to address that deficit by offering a system that includes a significant set of resources for audio applications. These include both hardware and software components.

Be sure to check out **[Volume 1: Introduction and Overview](#)**

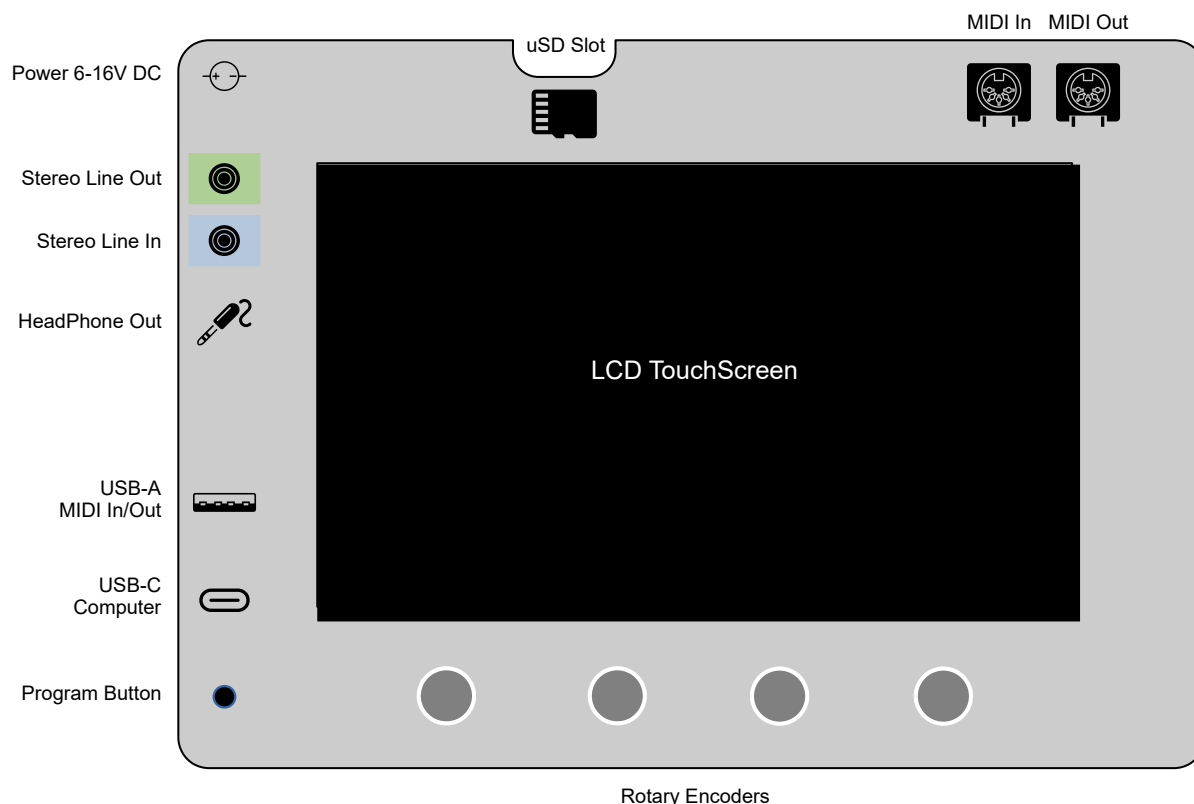
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# What's In the Box?

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## AudioPlatform Unit

The AudioPlatform unit features a matte black acrylic top panel, with the name appearing in vertical copper-colored lettering on the upper left (as seen in the cover image of this document). All input and output connectors are located on the edges of the AudioPlatform PCB, underneath the top panel, as depicted in the functional image below.



## AC/DC Adapter



AudioPlatform is shipped with a 7.5V AC/DC adapter connects to the unit with a center-positive (+), 2.1mm female connector, and conveys power to the unit when plugged into a standard AC outlet.

## microSD Card and Adapter



AudioPlatform ships 32GB Sandisk microSD card (inserted in the Teensy 4.1 microSD slot) and a MicroSD to SD adapter for use when a standard SD connection is required. The microSD card plays an integral role in the AudioPlatform software application (***apApp***), hosting a number of audio and data file types used by the application.

In particular, the microSD card shipped with AudioPlatform contains all the media and data files required by *apApp* in the examples and demonstration of application features described in this guide.

## Tilting Base



The tilting base stand holds AudioPlatform vertically or horizontally, in 5 lockable positions. The stand has a non-slip rubber base, and folds flat (5.91 x 5.02 x 0.47 inches) for easy transport.

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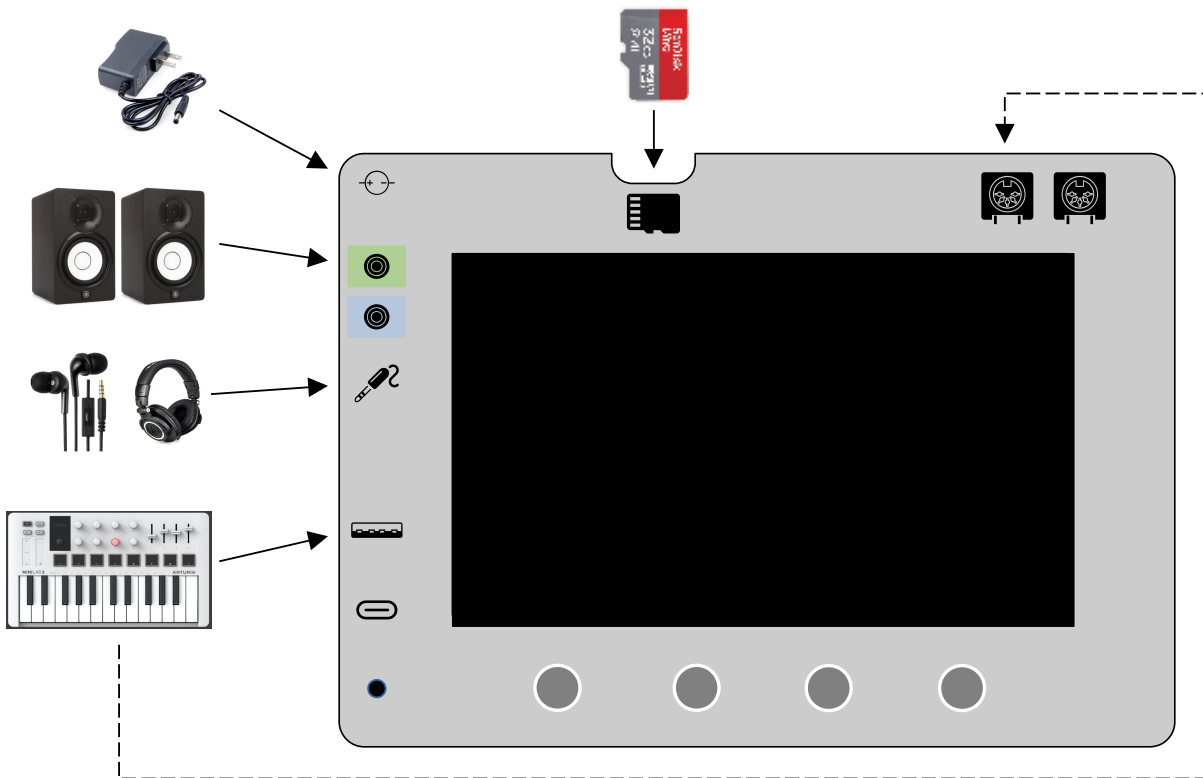
# Setup and Verification

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## Input/Output Connections

At a minimum, you'll need to connect the AC/DC adapter, speakers or headphones, and a MIDI controller that can generate MIDI note messages, as depicted below

In order to more quickly familiarize users with AudioPlatform, not all of the possible input and output configurations will be considered in this document. The aim is to get the basic system up and running, and the user comfortable with the basic functions of the pre-installed *apApp* AudioPlatform application. The minimal basic connections are depicted in the figure below:



Detailed descriptions of all aspects of the *apApp* application – screens, navigation, players, effects, system components, device settings, presets, system utilities and other operations are provided in the separate document *apApp User Guide* (see AudioPlatform Source and Documentation in the Appendix).

## microSD Card

Normally, the microSD card shipped with AudioPlatform will already be installed in the unit. However, if the microSD card is not present in the Teensy 4.1 microSD slot located in the cutout at the top of AudioPlatform, you should insert the card before applying power and starting the application.

## Power

Power is applied to AudioPlatform when the included AC/DC adapter is connected to the power input on the upper left side of the unit. It's generally OK to connect the power if the adapter is already plugged into a live AC connection, but it's also a good practice in audio to delay power up until all inputs and outputs are connected.

If your adapter is plugged in to a power strip with an on/off switch, then there is a means of turning AudioPlatform on and off using the switch on the strip. But this would potentially be a problem if other devices were connected to the same strip.

Another option is to use an in-line switch connected between the AC/DC adapter and AudioPlatform, so that AudioPlatform can be powered on and off independently:



This item is generally offered as an option when purchasing AudioPlatform, but can be purchased separately at <https://protosupplies.com/product/dc-jack-in-line-switch/> or other sources.

## Audio Input and Output



It's important to note that AudioPlatform's audio line and headphone outputs cannot drive non-powered speakers directly. In order to use non-powered speakers, AudioPlatform's outputs should be connected either to an audio mixer which itself provides the amplification required by the speaker, or that a suitable amplifier is connected between the mixer and the speaker.

## Input and Output Connectors

All audio input and output connections to AudioPlatform require stereo signals terminated in a 3.5mm TRS (Tip-Ring-Sleeve) connector, as shown in the image below.



## Input and Output Levels

Input and output levels are set on the Audio IO screen, accessible from the main menu of *apApp*.

## Audio Output

### Speakers

There are a great number of speaker types, from desktop PC speakers to professional studio monitors. However, one feature that divides the subject into two distinct categories is whether the speakers are self-powered, or instead require an amplifier to function. So, there is no single guideline for connecting speakers to AudioPlatform that is valid for all speakers.

Desktop PC speakers are typically both self-powered (generally by a USB power source) and feature a 3.5mm TRS stereo male connector for their input signal. For example:



USB-Powered speakers **cannot** be powered by AudioPlatform's USB-A connection. The USB-A port is limited to a current output of 500mA (per the USB specification). Powered speakers, even of low wattage, require currents in excess of this amount. For example, a single 8W speaker powered by a 5V connection will require about 1.5A. Therefore, USB-Powered speakers must be powered by a separate source.

Other powered speakers may require a converter cable that converts AudioPlatform's stereo 3.5mm stereo output to a pair of mono cables matching the speaker's input sockets. For example, the splitter cable below converts the 3.5mm stereo output to a pair of mono, 1/4" plugs compatible with a pair of speakers that are powered by a standard AC wall-cable.



In short, there are numerous types and configurations of speakers that users may wish to use with AudioPlatform, and the information provided here is intended only as a basic guide.



## Headphones

Most all headphones with a 3.5mm stereo male connector will work with AudioPlatform. Simply connect the headphones to the headphone output jack. Headphone output levels can be set, as mentioned above, on the Audio IO screen.

## Audio Input

At this stage, the intent is to familiarize users with the basics of AudioPlatform's audio processing. As such, we will only briefly describe the requirements for using audio input to AudioPlatform.

### Audio Mixer

As with the stereo line and headphone outputs, audio input is also a line-level stereo input. This means that, in order to use a microphone or other low-impedance devices, as well as using multiple audio sources, the best option is to use an audio mixer for connection of these sorts of audio sources to the AudioPlatform's line input. The following diagram illustrates the basic setup:



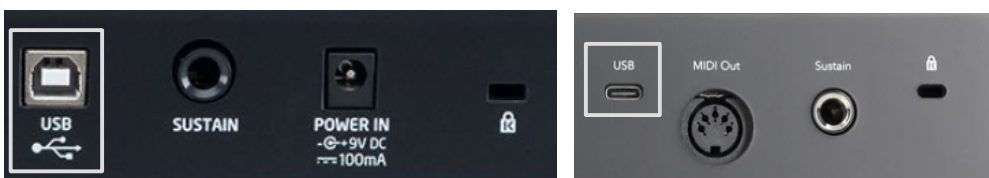
## MIDI Controller

In order to complete the setup and verification process, you will need a MIDI keyboard controller that is capable of sending MIDI note, and, optionally, modulation wheel, and pitch bend messages at its output. Whereas there are scores of such devices currently on offer, most every one will include these features. So, in the most basic configuration, a MIDI keyboard controller is required to verify and explore the features of the *apApp* software.

MIDI keyboard controllers come in a number of configurations and output types. For present purposes, the most important feature is the type of MIDI output connector, and the source of the controller's power requirements.

### USB MIDI

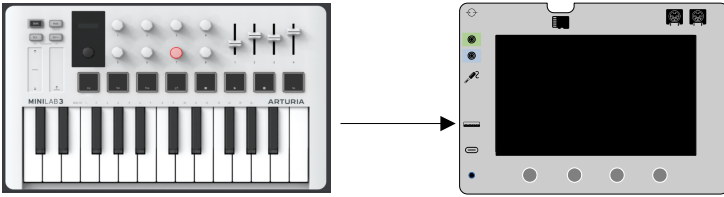
For some time now, MIDI controllers have featured USB output, as shown in the images below:





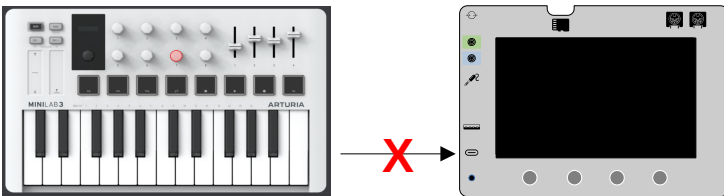
## USB-A Connector

AudioPlatform's USB-A connector is used exclusively for connection of a MIDI controller, and can supply power – up to a maximum of 500mA – to a connected MIDI device.



## USB-C Connector

The USB-C connection is used exclusively for connection of AudioPlatform to a computer for programming under the Arduino IDE. AudioPlatform's USB-C computer connection cannot supply power to a connected MIDI device.



## Legacy DIN MIDI

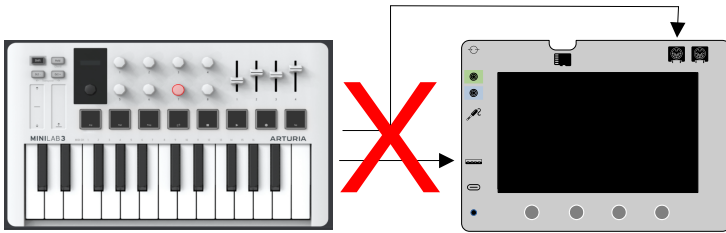
More recent MIDI controllers, in addition to USB output, generally include a round 5-pin “legacy” MIDI DIN output in support of the numerous pre-USB MIDI devices that many users still employ. This arrangement is depicted below. This device provides MIDI output on both the USB connector and the legacy DIN connector, and draws power from the USB connection.



To use a legacy device with AudioPlatform, the device's DIN output is connected to AudioPlatform's MIDI DIN input. In this case, the controller is powered by an AC-to-USB power converter:



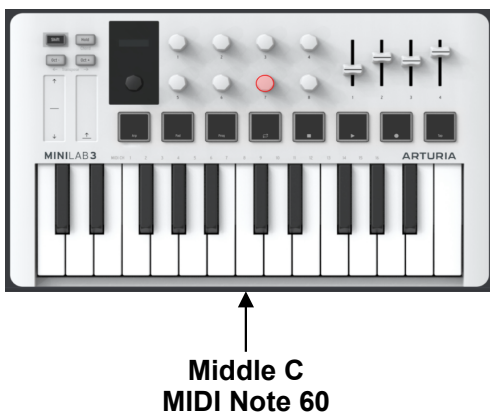
Although AudioPlatform can supply power from its USB-A connection, the USB-A should not be used to power a legacy MIDI device. Although this arrangement might (or might not) succeed in supplying power to the legacy device, the result would most likely be that MIDI messages would be received by AudioPlatform on both its MIDI DIN and USB-A ports, resulting in duplicate MIDI messages.



## apApp Keyboard Mapping in Demonstration Presets

In general, regardless of whether a MIDI keyboard has 25, 29, 61 or more keys, the default setting for most keyboards places MIDI note 60 in the middle of the keyboard. For keyboards with a smaller number of keys, buttons are provided to shift the keyboard range up and down in order to cover the full range of MIDI notes, 0-127.

Therefore, in order to simplify the process of getting acquainted with AudioPlatform, all demonstration sounds and presets are “mapped” to the 2-octave (25 note) range of MIDI notes 36 to 72, with MIDI note 60 (middle-C) in the center:

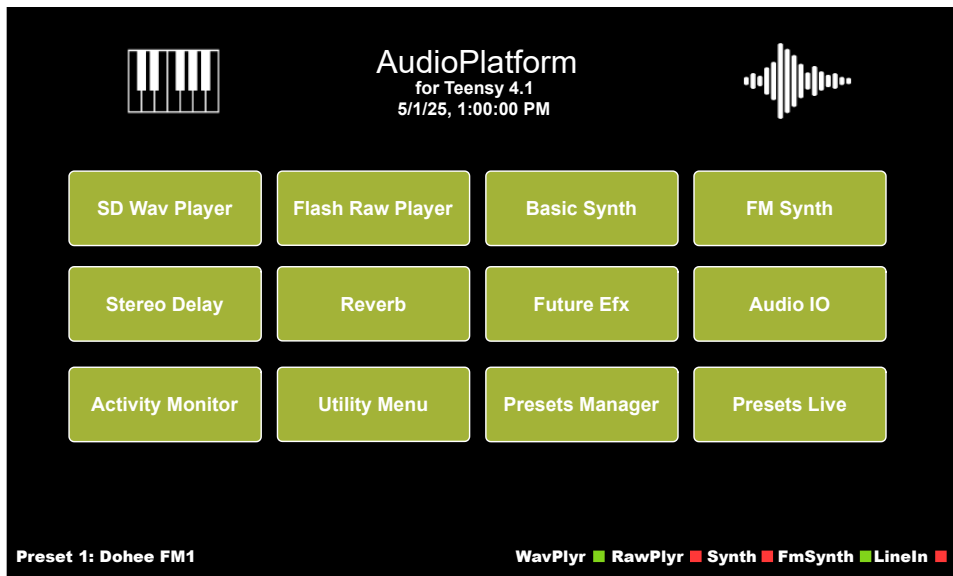


This arrangement ensures that users will be able to audition the sounds and presets with the least possible adjustments to the MIDI controller.

# Verification

## Power On

Once power is connected to AudioPlatform, the Main Menu screen should appear after a few seconds:



If the Main Menu fails to appear, first verify that the unit is receiving power, in which case a small green LED, located on the PCB baseboard below the power input, will be turned on.

## Screens and Buttons

Apart from the MIDI controller and AudioPlatform's rotary encoders, user interactions with *apApp* are managed via the LCD touchscreen. The Main Menu features 9 *buttons* which, when touched, cause various other interactive *screens* to be displayed. These buttons take the form of the olive green buttons of the main menu and generally result in the display of a new screen.



Other icons of various colors and function also aid the user in navigation among screens and pop-up items:



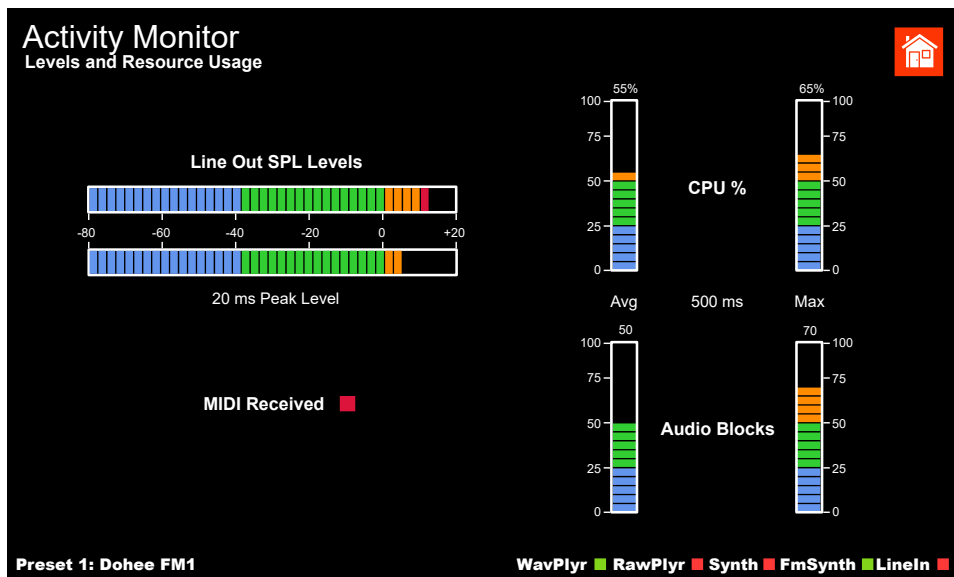
## MIDI Input and Audio Output

If everything – Power, MIDI Controller, Audio Out / Speakers / Headphones – is connected properly, try pressing Middle-C on the keyboard. If all is correct, you should hear a familiar tubular bell sound (likely familiar to anyone who has seen a Taco Bell commercial, where this sound has been used since the days of the original DX7 synthesizer).

If you're not hearing any sound, you can use the Activity Monitor screen to verify that MIDI input is received and sound is being generated. The screen is accessed from the Main Menu by touching the screen's button:



You should then see the following screen:



If MIDI input is detected, the red “LED” to the right of the “MIDI Received” text should turn green for a short time. If the LED remains red, you’ll need to re-check the connections of your MIDI controller to AudioPlatform.

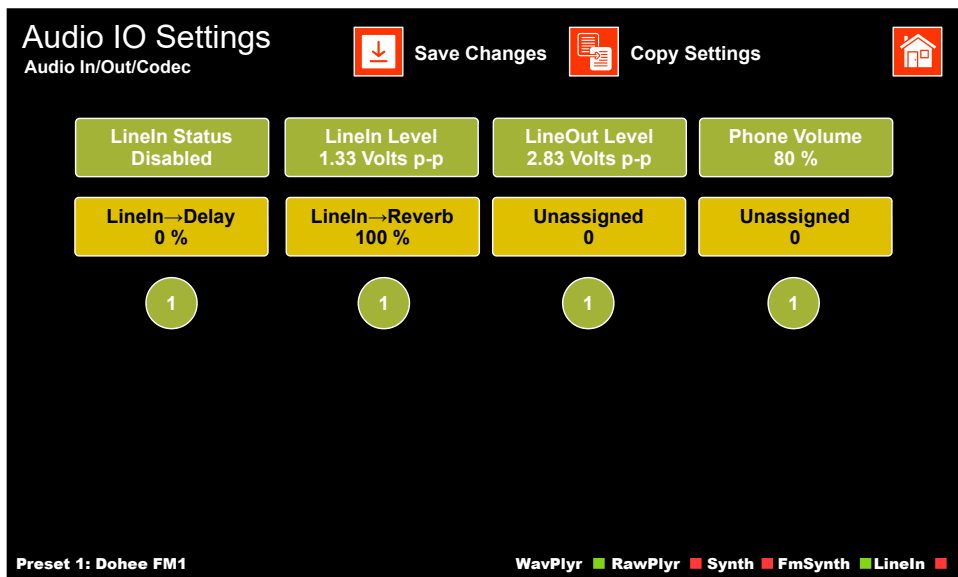
If audio output is being generated correctly when a key is pressed, you should see the stereo output levels displayed on the “Line Out SPL Levels” bars. If you see no change in the levels, then there is no sound being generated, most likely because no MIDI input is being received.

If you can see the changes to the stereo output levels, but can’t hear any sound, you’ll need to first check to ensure that your audio output is properly set up. This is most easily accomplished by connecting a set of headphones to AudioPlatform’s headphone output jack.

If you still hear no sound in your headphones, activate the Audio Settings screen by pressing the Main Menu button:



which will display the Audio IO settings screen:



Make sure that the Phone Volume is turned up to a level sufficient to be heard in your headphones (the level should be set to 60% by default; but, if not, adjust the volume to a suitable level).

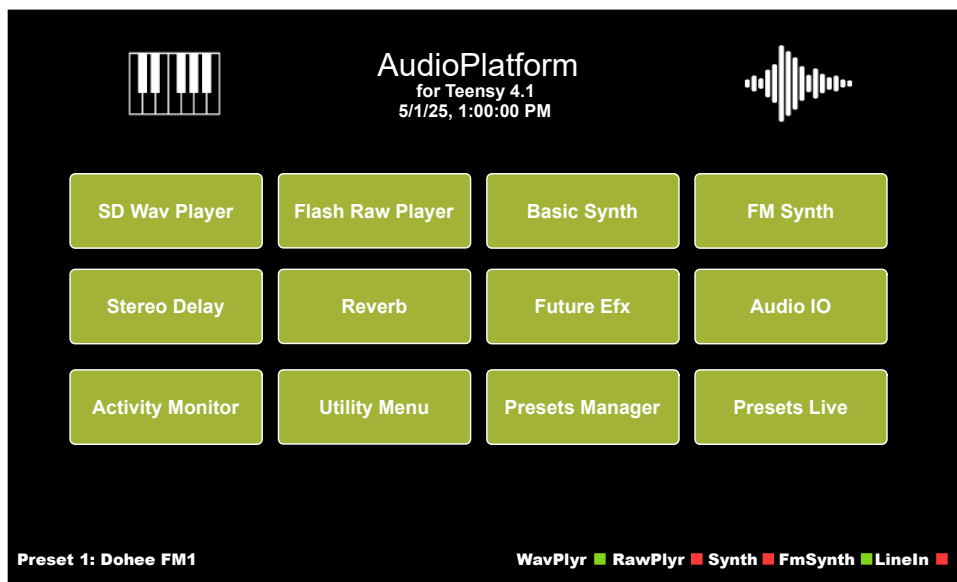
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## Quick Overview

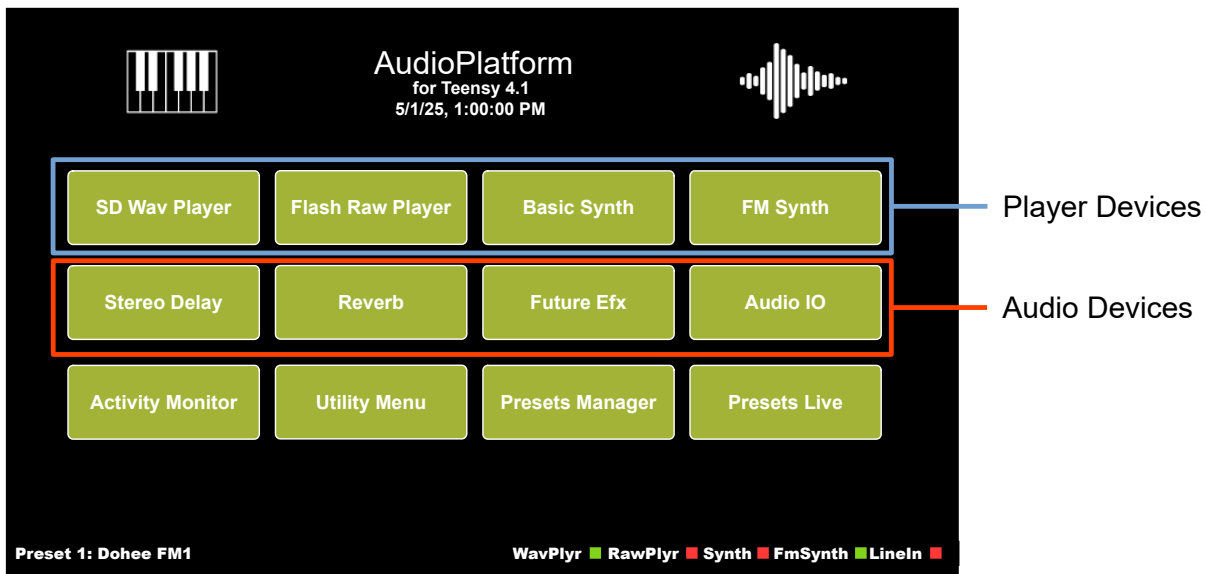
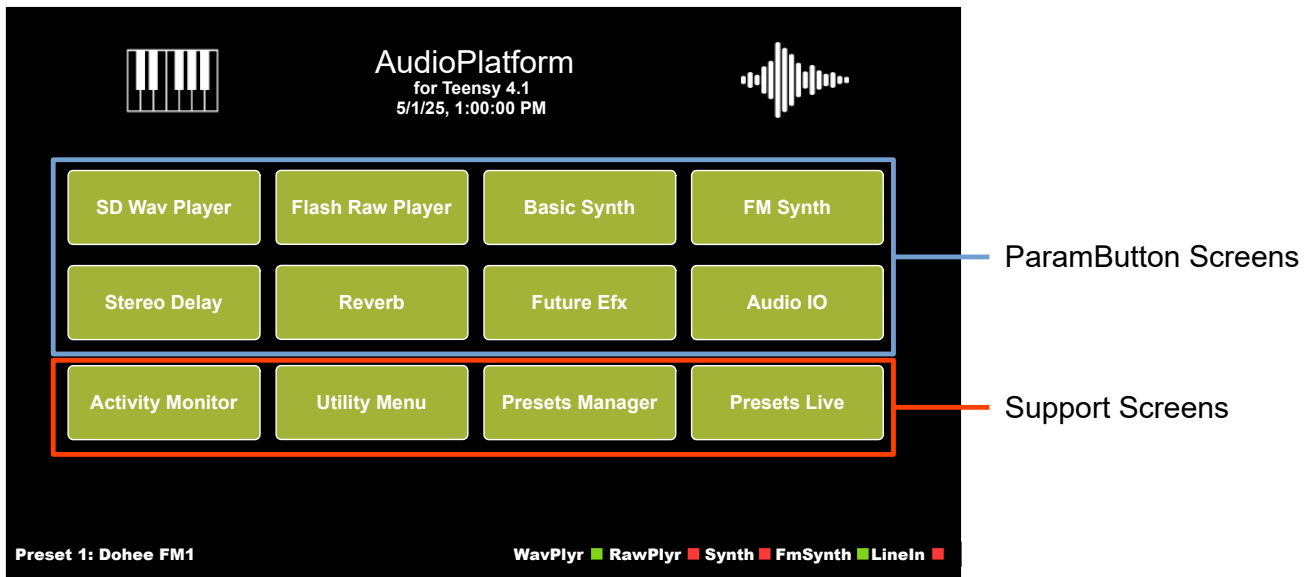
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### Startup Screen (Main Menu)

On startup, the *Main Menu* screen will be displayed.

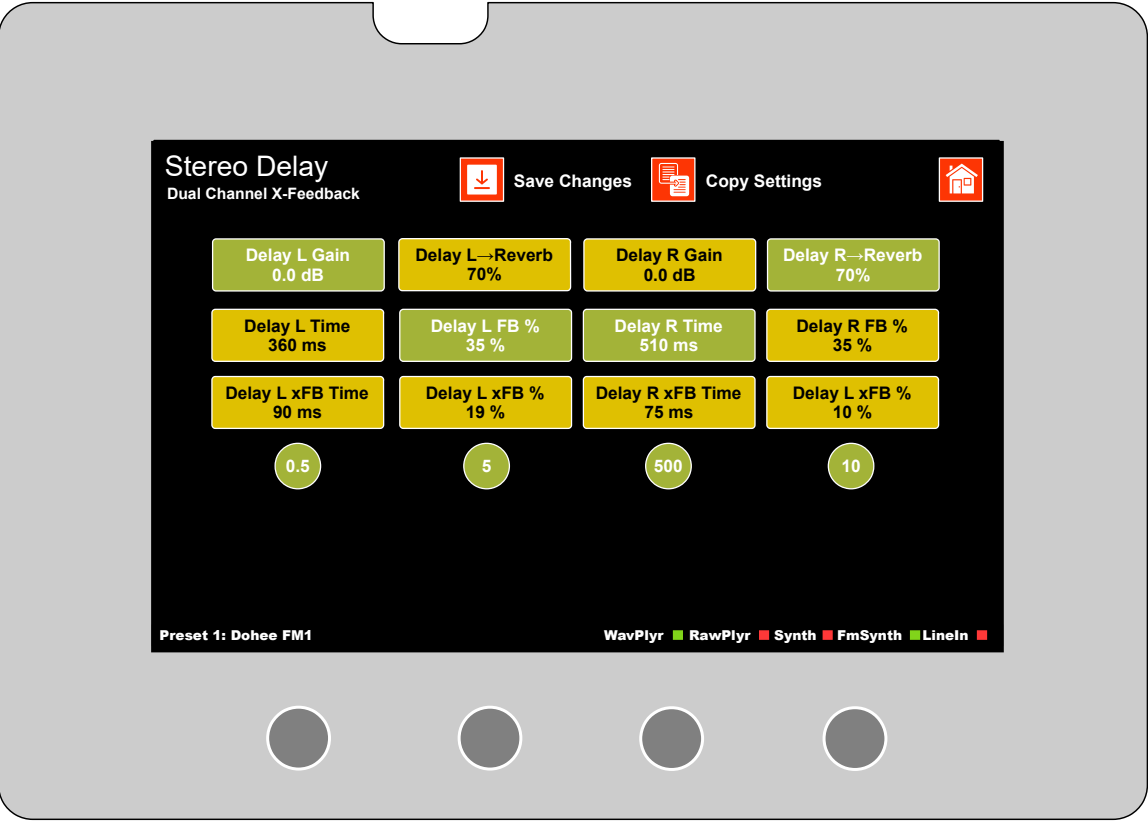


## Main Menu Screen Types





# ParamButton Screens



- 0.5, 1, 6, 12
- 1, 5, 10
- 1, 5, 10, 50, 100, 500
- 1, 5, 10

## Navigation



Home



Back



Enter



Exit

## Home



This button appears in the upper-right corner of all level 2 screens (screens one level below the main menu). Touching the button returns the user to the Main Menu screen.

## Back



This button appears in the upper-right corner of level 3 and greater screens (screens at least two levels below the Main Screen).

## Enter



This button may appear in various screens to provide access to a lower-level screen that is only accessible from the current screen.

## Exit



This button appears in screens accessed using the Enter button, allowing the user to return to the screen which invoked the currently displayed screen.

## Player Device Screens

### Player Device Common Settings

- **Player Status**
- **Player Level**
- **Player → Delay**
- **Player → Reverb**
- **Note Shift**
- **Note Min**
- **Note Max**
- **Pan Type**
- **Pan Parameter**



**Middle C**  
**MIDI Note 60**

## WAV Player

### WAV File Player

Teensy4.1 SD WAV Files

 Save Changes  Copy Settings 

|                          |                        |                      |                      |
|--------------------------|------------------------|----------------------|----------------------|
| Player Status<br>Enabled | Player Level<br>0.0 dB | Player→Delay<br>58 % | Player→Reverb<br>80% |
| Sound Bank<br>0          | Note Shift<br>-12      | Note Min<br>0        | Note Max<br>99       |
| Pan Type<br>Center       | Pan Parameter<br>63.0  | Unassigned<br>0      | Unassigned<br>0      |
| 1                        | 0.5                    | 1                    | 1                    |

Preset 1: Dohee FM1

WavPlyr  RawPlyr  Synth  FmSynth  LineIn 

## RAW Player

### Raw File Player

Serial Flash Raw Audio Files

 Save Changes  Copy Settings 

|                           |                         |                     |                     |
|---------------------------|-------------------------|---------------------|---------------------|
| Player Status<br>Disabled | Player Level<br>-9.5 dB | Player→Delay<br>0 % | Player→Reverb<br>0% |
| Sound Bank<br>0           | Note Shift<br>0         | Note Min<br>0       | Note Max<br>99      |
| Pan Type<br>Center        | Pan Parameter<br>63.0   | Unassigned<br>0     | Unassigned<br>0     |
| 1                         | 0.5                     | 1                   | 1                   |

Preset 1: Dohee FM1

WavPlyr  RawPlyr  Synth  FmSynth  LineIn 

## Basic Synthesizer

Basic Synthesizer  
12 Voice Polyphony

 Save Changes  Copy Settings 

|                          |                       |                     |                       |
|--------------------------|-----------------------|---------------------|-----------------------|
| Synth Status<br>Disabled | Synth Level<br>0.0 dB | Player→Delay<br>0 % | Player→Reverb<br>0%   |
| Synth Type<br>AM         | Note Shift<br>0       | Note Min<br>0       | Note Max<br>99        |
| Attack<br>0 ms           | Decay<br>0 ms         | Sustain<br>0 %      | Release<br>0 ms       |
| Mod Freq<br>0 Hz         | Mod Amt<br>80 %       | Pan Type<br>Center  | Pan Parameter<br>63.0 |
| 1                        | 0.5                   | 1                   | 1                     |

Preset 1: Dohee FM1

WavPlyr  RawPlyr  Synth  FmSynth  Lineln 

## FM Synthesizer

### Basic Synthesizer

12 Voice Polyphony

 Save Changes  Copy Settings 

Synth Status  
Disabled

Synth Level  
-8.5 dB

Player→Delay  
20 %

Player→Reverb  
60%

 Voices

Voice Name  
Lute

Note Shift  
14

Note Min  
0

Note Max  
127

PitchBend Range  
1 Semitones

Pan Type  
Center

Pan Parameter  
63.0

Unassigned

1

0.5

1

1

Preset 1: Dohee FM1

WavPlyr  RawPlyr  Synth  FmSynth  Lineln 

## FM Synthesizer Voice Banks

### FM Synth Voice Banks

Select Bank and Voice

Voice

|            |           |            |           |             |            |           |          |
|------------|-----------|------------|-----------|-------------|------------|-----------|----------|
| BRASS 1    | BRASS 2   | BRASS 3    | STRINGS 1 | STRINGS 2   | STRINGS 3  | ORCHESTRA | PIANO 1  |
| PIANO 2    | PIANO 3   | E.PIANO 1  | GUITAR 1  | GUITAR 2    | SYN.LEAD 1 | BASS 1    | BASS 2   |
| E.ORGAN 1  | PIPES 1   | HARPSICH 1 | CLAV 1    | VIBE 1      | MARIMBA    | KOTO      | FLUTE 1  |
| ORCH-CHIME | TUB BELLS | STEEL DRUM | TIMPANI   | REFS WHISTL | VOICE 1    | TRAIN     | TAKE OFF |

Voice Bank

|        |        |       |       |       |       |       |       |
|--------|--------|-------|-------|-------|-------|-------|-------|
| rom1a  | rom1b  | rom2a | rom2b | rom3a | rom3b | rom4a | rom4b |
| Ritch1 | Ritch2 | User1 | User2 | User3 | User4 | User5 | User6 |

  
FM Synth

## FM Synthesizer Emulator

synth\_dexed is a port of the Dexed sound engine (<https://github.com/asb2m10/dexed>) as library for the Teensy-3.5/3.6/4.x with an audio shield.

Dexed ist heavily based on <https://github.com/google/music-synthesizer-for-android>

synth\_dexed.cpp/h used in AudioPlatform is from:

Arduino Resources\Third Party Code\microdexed-touch\third-party\Synth\_Dexed\src\

## PC/Mac Editor

<https://asb2m10.github.io/dexed/>



## Audio Device Screens

### Stereo Delay

#### Audio IO Settings

Audio In/Out/Codec

 Save Changes  Copy Settings 

|                           |                                |                                 |                      |
|---------------------------|--------------------------------|---------------------------------|----------------------|
| LineIn Status<br>Disabled | LineIn Level<br>1.33 Volts p-p | LineOut Level<br>2.83 Volts p-p | Phone Volume<br>80 % |
| LineIn→Delay<br>0 %       | LineIn→Reverb<br>100 %         | Unassigned<br>0                 | Unassigned<br>0      |
| 1                         | 1                              | 1                               | 1                    |

Preset 1: Dohee FM1

WavPlyr  RawPlyr  Synth  FmSynth  LineIn 

### Reverb and Special EFX

#### Effects Settings

Reverb and Future Fx

 Save Changes  Copy Settings 

|                   |                          |                        |                 |
|-------------------|--------------------------|------------------------|-----------------|
| Reverb Size<br>80 | Reverb Diffusion<br>70 % | Reverb Level<br>0.0 dB | UnAssigned<br>0 |
| Unassigned<br>0   | Unassigned<br>0          | Unassigned<br>0        | Unassigned<br>0 |
| 0.5               | 1                        | 0.5                    | 1               |

Preset 1: Dohee FM1

WavPlyr  RawPlyr  Synth  FmSynth  LineIn 



## Audio IO

### Audio IO Settings

Audio In/Out/Codec

 Save Changes  Copy Settings 

|                           |                                |                                 |                      |
|---------------------------|--------------------------------|---------------------------------|----------------------|
| LineIn Status<br>Disabled | LineIn Level<br>1.33 Volts p-p | LineOut Level<br>2.83 Volts p-p | Phone Volume<br>80 % |
| LineIn→Delay<br>0 %       | LineIn→Reverb<br>100 %         | Unassigned<br>0                 | Unassigned<br>0      |
| 1                         | 1                              | 1                               | 1                    |

Preset 1: Dohee FM1 WavPlyr  RawPlyr  Synth  FmSynth  LineIn 

## MIDI Settings and Controllers

### MIDI Settings

Future Development

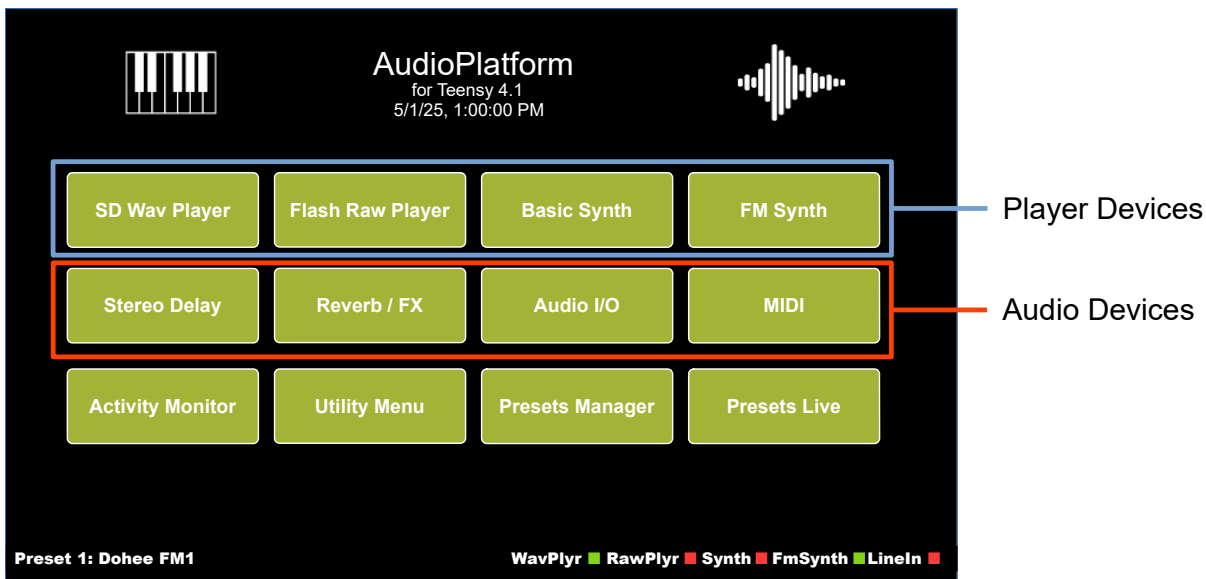
 Save Changes  Copy Settings 

|                       |                               |                      |                      |
|-----------------------|-------------------------------|----------------------|----------------------|
| MIDI Port<br>USB MIDI | MIDI In Device<br>LaunchKey25 | Wav Player Chan<br>1 | Raw Player Chan<br>1 |
| BasicSynth Chan<br>1  | FmSynth Chan<br>1             | Unassigned<br>0      | Unassigned<br>0      |
| Unassigned<br>0       | Unassigned<br>0               | Unassigned<br>0      | Unassigned<br>0      |
| Unassigned<br>0       | Unassigned<br>0               | Unassigned<br>0      | Unassigned<br>0      |
| 1                     | 1                             | 1                    | 1                    |

Preset 1: Dohee FM1 WavPlyr  RawPlyr  Synth  FmSynth  LineIn 

## Players

# Player and Audio Devices



## Player Devices

*Player Devices* occupy the topmost row of the Main Menu screen. apApp features 4 players, which play audio media in response to inputs from an external MIDI device, and each of which operates on a separate media type:

- The *SD Wav Player* plays Stereo WAV formatted files located on the Teensy 4.1 SD card
- The *Flash Raw Player* plays Monaural (single channel, headerless) audio files located in the AudioPlatform's 128 MB non-volatile Flash Memory
- The *Basic Synth* is an implementation of a basic digital audio synthesizer with 12-voice polyphony
- The *FM Synth* is a Teensy Audio System implementation of the micro-Dexed synthesizer which is an emulator of the classic DX7 synthesizer, featuring 8-voice polyphony

## Audio Devices

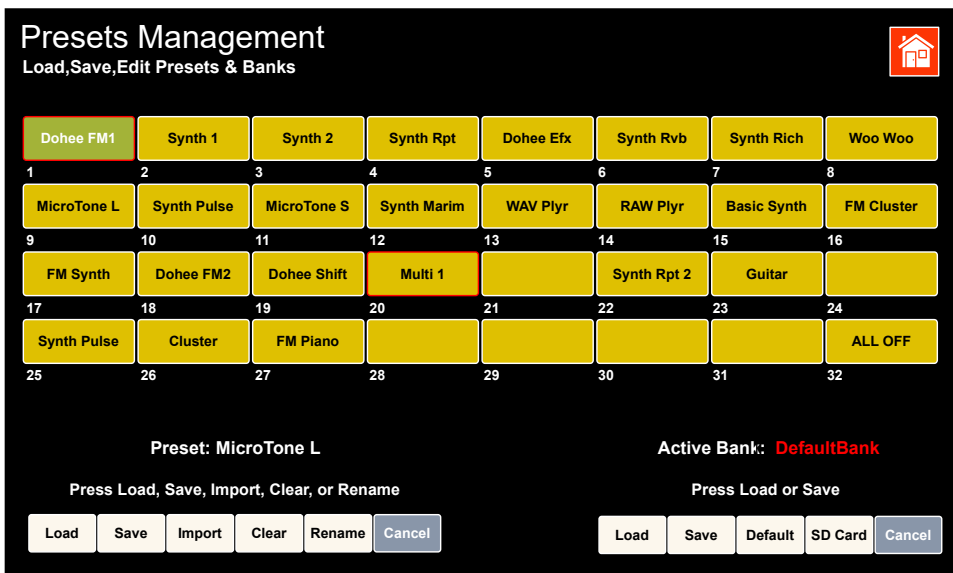
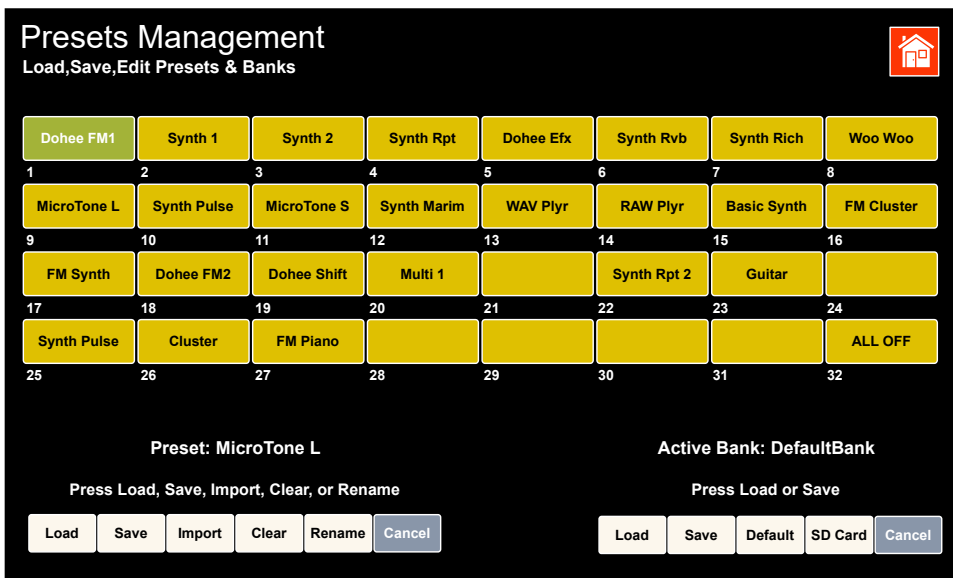
*Audio Devices* occupy the middle row of the Main Menu screen. apApp features 4 audio devices, which play various roles in the apApp audio architecture:

- The *Stereo Delay* device implements a stereo “ping-pong” digital delay, with up to 3 seconds of delay per channel, and both individual channel and cross-channel feedback paths
- The *Reverb / EFX* device features a Plate Reverberation unit, and a second effects unit pathway intended for addition of a 3<sup>rd</sup> audio effect in future apApp releases.

- The *Audio IO* device provides for control of the levels and effects settings of the system audio paths, with settings for Input and Output levels, effects send levels, and other audio system parameters
- The *MIDI* device features settings related to a connected MIDI control device, such as MIDI channel and destination player settings

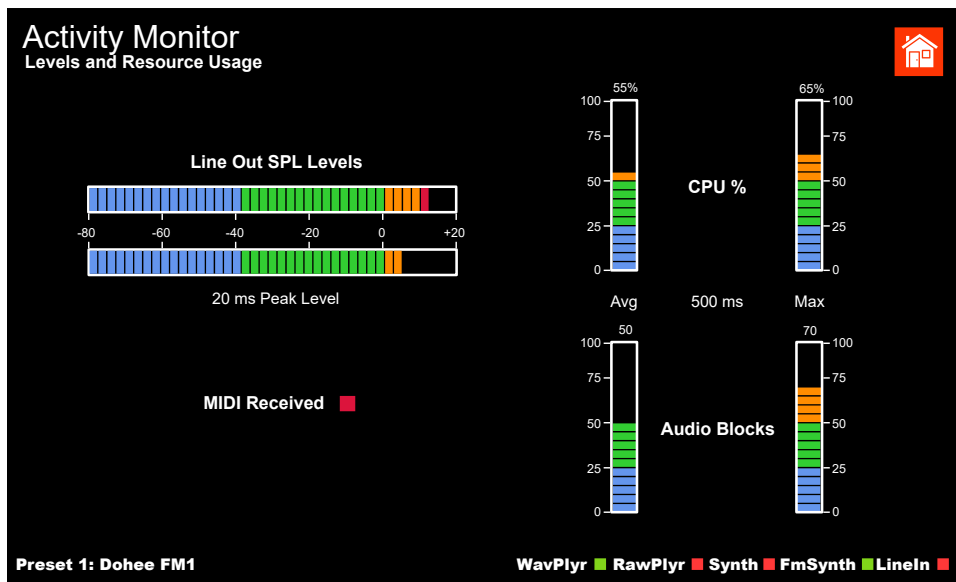
## Efx and MIDI Devices

### Presets Manager

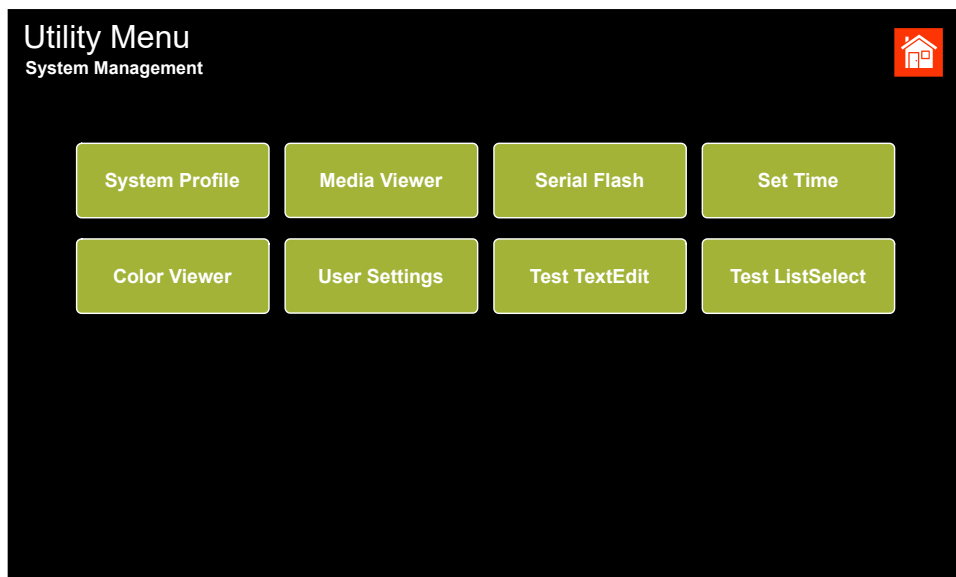




## Activity Monitor



## Utility Menu



## Presets and Parameters

# Screens and Menus

## Keyboard Mapping

### microSD Card

Although the AudioPlatform application can be operated without the presence of a microSD card (generally referred to in this document as the SD Card) in the Teensy 4.1 card slot, there are a number of features of the system that are dependent on the presence of the card and its contents. These include:

- Access to Stereo WAV (".wav") audio files located on the SD card for playback by the SD Wav Player
- Access to banks of Monaural (Mono) RAW (".raw") audio files to be transferred to the internal Serial Flash storage for playback by the Flash Raw Player
- Access to user Preset Banks (".pbnk") files stored on the card
- Access to user-defined Voice Banks (".syx") for playback using the FM Synthesizer

Cards with "A1", "A2" or higher ratings recommended for good performance

## Presets Live

### Presets Live

Direct Preset & Bank Access

All Notes Off
 Ignore Note Off

|                |                |                |                |                                                                                                                                                                                                                                                                                                                         |
|----------------|----------------|----------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 Dohee FM1    | 2 Synth 1      | 3 Synth 2      | 4 Synth Rpt    | <b>DefaultBank.pbk</b><br><br>DefaultBank-250220.pbk<br>DefaultBank-250319.pbk<br>DefaultBank-250415.pbk<br>DefaultBank-250519.pbk<br>ImportActiveTest.pbk<br>SynthMicroToneCmnDly.pbk<br>FmSynthPanTest.pbk<br>MultiPlayerTest1.pbk<br>MultiPlayerTest2.pbk<br>WavPlayerMultiEffects.pbk<br>BasicSynthMultiEffects.pbk |
| 5 Dohee Efx    | 6 Synth Rvb    | 7 Synth Rich   | 8 Woo Woo      |                                                                                                                                                                                                                                                                                                                         |
| 9 MicroTone L  | 10 Synth Pulse | 11 MicroTone S | 12 Synth Marim |                                                                                                                                                                                                                                                                                                                         |
| 13 WAV Plyr    | 14 Raw Plyr    | 15 Basic Synth | 16 FM Cluster  |                                                                                                                                                                                                                                                                                                                         |
| 17 FM Synth    | 18 Dohee FM2   | 19 Dohee Shift | 20 Multi 1     |                                                                                                                                                                                                                                                                                                                         |
| 21             | 22 Synth Rpt 2 | 23 Guitar      | 24             |                                                                                                                                                                                                                                                                                                                         |
| 25 Synth Pulse | 26 Cluster     | 27 FM Piano    | 28             |                                                                                                                                                                                                                                                                                                                         |
| 29             | 30             | 31             | 32 ALL OFF     |                                                                                                                                                                                                                                                                                                                         |

Active Bank: DefaultBank.pbk

Preset Banks on SD

Preset 1: Dohee FM1

WavPlyr ■ RawPlyr ■ Synth ■ FmSynth ■ LineIn ■

## microSD Contents

### Stereo WAV Files

#### Bank 0

000-SoundFile.wav  
001-SoundFile.wav  
002-SoundFile.wav  
003-SoundFile.wav  
004-SoundFile.wav  
005-SoundFile.wav  
006-SoundFile.wav  
007-SoundFile.wav  
008-SoundFile.wav  
009-SoundFile.wav  
010-SoundFile.wav  
011-SoundFile.wav  
012-SoundFile.wav  
013-SoundFile.wav  
014-SoundFile.wav  
015-SoundFile.wav  
016-SoundFile.wav  
017-SoundFile.wav  
018-SoundFile.wav  
019-SoundFile.wav  
020-SoundFile.wav  
021-SoundFile.wav  
022-SoundFile.wav  
023-SoundFile.wav  
024-SoundFile.wav

#### Bank 1

100-SoundFile.wav  
101-SoundFile.wav  
102-SoundFile.wav  
103-SoundFile.wav  
104-SoundFile.wav  
105-SoundFile.wav  
106-SoundFile.wav  
107-SoundFile.wav  
108-SoundFile.wav  
109-SoundFile.wav  
110-SoundFile.wav  
111-SoundFile.wav  
112-SoundFile.wav  
113-SoundFile.wav  
114-SoundFile.wav  
115-SoundFile.wav  
116-SoundFile.wav  
117-SoundFile.wav  
118-SoundFile.wav  
119-SoundFile.wav  
120-SoundFile.wav  
121-SoundFile.wav  
122-SoundFile.wav  
123-SoundFile.wav  
124-SoundFile.wav



## **Stereo WAV Files (continued)**

### **Bank 2**

200-  
201-  
202-  
203-  
204-  
205-  
206-  
207-  
208-  
209-  
210-  
211-  
212-  
213-  
214-  
215-  
216-  
217-  
218-  
219-  
220-  
221-  
222-  
223-  
224-

### **Bank 3**

300-  
301-  
302-  
303-  
304-  
305-  
306-  
307-  
308-  
309-  
310-  
311-  
312-  
313-  
314-  
315-  
316-  
317-  
318-  
319-  
320-  
321-  
322-  
323-  
324-

## Mono RAW Files

### Bank 0

000-  
001-  
002-  
003-  
004-  
005-  
006-  
007-  
008-  
009-  
010-  
011-  
012-  
013-  
014-  
015-  
016-  
017-  
018-  
019-  
020-  
021-  
022-  
023-  
024-

### Bank 1

100-  
101-  
102-  
103-  
104-  
105-  
106-  
107-  
108-  
109-  
110-  
111-  
112-  
113-  
114-  
115-  
116-  
117-  
118-  
119-  
120-  
121-  
122-  
123-  
124-

## ***Mono RAW Files (continued)***

### **Bank 2**

200-  
201-  
202-  
203-  
204-  
205-  
206-  
207-  
208-  
209-  
210-  
211-  
212-  
213-  
214-  
215-  
216-  
217-  
218-  
219-  
220-  
221-  
222-  
223-  
224-

### **Bank 3**

300-  
301-  
302-  
303-  
304-  
305-  
306-  
307-  
308-  
309-  
310-  
311-  
312-  
313-  
314-  
315-  
316-  
317-  
318-  
319-  
320-  
321-  
322-  
323-  
324-

## **SD Card Usage and Management**

Although the presence of an SD card is not required to use many of the basic features of apApp, the full extent of application capabilities is only possible when an SD card is present in the Teensy 4.1 microSD slot.

Because of this and various other considerations, it is considered fundamental to the full benefits of AudioPlatform that the contents of the SD card be managed in a well-organized and careful manner.

Other than the capability to name, store, and load preset banks to/from the sdCard, apApp makes no provision of sdCard file and directory management. It is thus necessary that users organize and manage soundfile resources and SD card contents on a PC (Windows, Mac, Linux) using various available applications and/or utilities. The renaming of a collection of audio files to conform to the naming requirements for SD card audio files in apApp is perhaps the primary issue in this connection,

and it is necessary to use some appropriate means of doing so. There are numerous paid and free applications that can be used for this purpose. Examples include:

## **PC-Based File Management Utilities**

### **Windows**

- Microsoft PowerToys PowerRename
- Advanced Renamer
- Bulk Rename Utility
- File Renamer
- Metamorphose 2

### **MacOS**

- Finder's Built-in Feature
- Transnomino
- NameChanger
- Metamorphose 2

### **Linux**

- GPRename
- pyRenamer
- Thunar
- Metamorphose 2

## **SD Card Format**

## **SD Card Content**

### **File Naming**

In order to be recognized in apApp, WAV and RAW audio filenames must follow a specific convention that indicates which of the 4 sample banks the file is to be located in, and which MIDI note number (0-99, 100-127 ignored) will play the sample.

The convention dictates that the sample filename must begin with a 3-digit number (leading 0s required) followed by a dash '-' and the filename, and ending with the suffix .wav or .raw as the case may be. The 3 digits indicate both the bank number:

- 000-099      Bank 0, MIDI notes 0-99

- 100-199 Bank 1, MIDI notes 0-99
- 200-299 Bank 2, MIDI notes 0-99
- 300-399 Bank 2 MIDI notes 0-99

## Examples

### **Bank 0**

- 000-Note0\_Sample.wav
- 060-MiddleC\_Sample.wav
- 099-Note99\_Sample.wav

### **Bank 2**

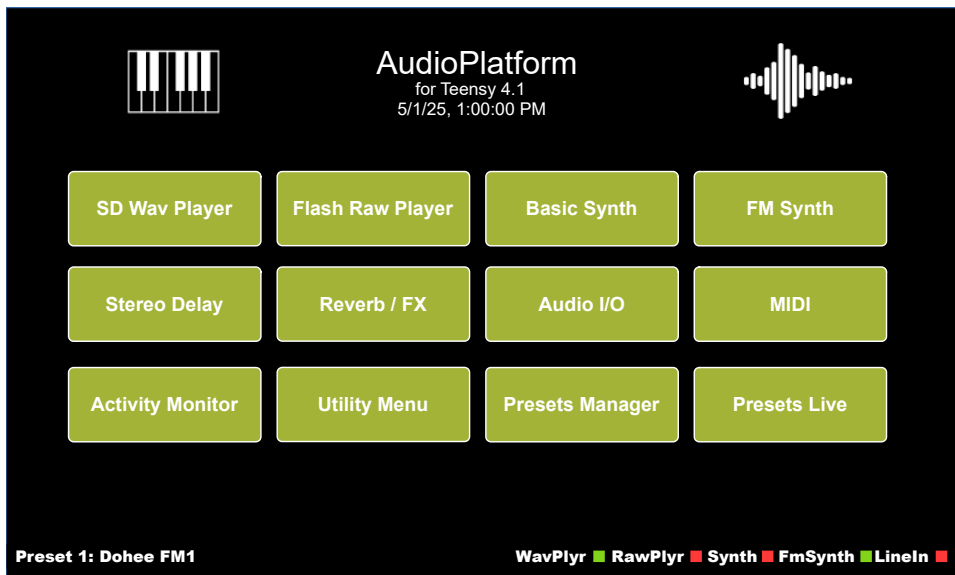
- 250-SoundEffect\_01.wav
- 260-SoundEffect\_02.wav
- 299-HighVoice.wav

The same conventions are used for RAW audio file banks stored in non-volatile Serial Flash.

## Filename Management

In general, the source of most user-supplied audio WAV and Raw files will be

# Main Menu



## Support and Utility Devices

*Support Devices* occupy the bottom row of the Main Menu screen, and serve as a gateway to additional screens and menus that perform various system and user support functions.

- The *Activity Monitor*
- The *Utility Menu*
- The *Presets Manager*
- The *Presets Live* device

---

# Screens and Settings

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## Screen Icons

Screen icons (*buttons*) appear on all screens, and serve to allow the user to navigate among application screens (*Navigation Icons*), and to make selections or perform actions on the currently displayed screen (*Action Icons*).

### Navigation Icons

#### Home



This button appears in the upper-right corner of all level 2 screens (screens one level below the main menu). Touching the button returns the user to the Main Menu screen.

#### Back



This button appears in the upper-right corner of level 3 and greater screens (screens at least two levels below the Main Screen).

#### Enter



This button may appear in various screens to provide access to a lower-level screen that is only accessible from the current screen.

#### Exit



This button appears in screens accessed using the Enter button, allowing the user to return to the screen which invoked the currently displayed screen.

### Action Icons

#### All Notes Off



This button appears at the top of the Presets Live screen. When pressed, all notes currently playing are silenced. (Note: this function can also be assigned to a button or other control on a connected MIDI keyboard or controller).

### Ignore Note Off



This button appears at the top of the Presets Live screen. When pressed, all notes currently held or played thereafter continue to play, ignoring key or sustain pedal releases. (Note: this function can also be assigned to a button or other control on a connected MIDI keyboard or controller).

### Save Changes



This button appears at the top of all ParamButton screens. The text label of the button “Save Changes” normally appears in white. If any parameter changes are made to the displayed device using a rotary encoder, the text changes to red “**Save Changes**”, and remains so until the button is touched, at which point it returns to its normal white appearance. This process is discussed in detail in the **SECTION REFERENCE**.

### Copy Settings



This button appears on all ParamButton screens next to the Save Changes button. When pressed, it displays a screen with buttons representing the presets currently loaded in the *Import* presets bank. When one of the import preset buttons is touched, the device settings of the import device are copied to the currently displayed device. This process is discussed in detail in the **SECTION REFERENCE**.

### Scroll Up Down



These buttons appear on screens that display lists of files or other names either for review or selection of items (for example, a preset bank file on the SD card). When simply displaying lists, the screen will scroll up or down repeatedly until the button is released. When displaying items for selection, repeated touches are required to scroll the list.





## Supprt Screens

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# Appendix

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## References

### **AudioPlatform Source and Documentation**

[AudioPlatform Source Code](#)

[AudioPlatform Documentation](#)

### **ProtoSupplies**

[ProtoSupplies Web Site](#)

[Prototyping System For Teensy 4.1](#)

[Teensy 4.1 Fully Loaded For Prototyping System](#)

[Project System For Teensy 4.1](#)

[Teensy 4.x Audio Adapter](#)

### **Audio and USB Gear**

[Audio Cable Selector](#)

[USB Cables](#)

[MIDI Controllers](#)

### **PJRC / Teensy**

[PJRC](#)

[Teensy User Forum](#)

[Teensy 4.1](#)

[TeensyDuino](#)

[TeensyDuino Libraries](#)

[Teensy Audio Shield](#)

[Teensy Audio Tutorial](#)

### **Teensy DIY Audio Projects and Kits**

[Teensy Forum](#)

[TeensyMix Synth - DIY 8 Voice Synthesizer](#)

[Shruti II Teensy 4.1 DIY Synthesizer](#)

[Polyhonic Teensy DIY Sampler](#)

[Teensy Super Audio Board](#)

[MicroDexed Touch Synthesizer](#)

[Polyhonic Teensy DIY Sampler](#)

## YouTube

[DIY Dirtywave M8 headless using a Teensy 4.1](#)

[DX7 Clone on Teensy 4.0](#)

[Electrotechnique TSynth - Teensy 4.1 based DIY Synthesizer](#)

[Teensy 4.1 Drone Synth](#)

[Teensy Midi Step Sequencer](#)

[FALA Synthesizer - 8 Voice DIY Teensy Synth](#)